

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8630**

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Block:409	Seat No:	2022700011		
Written	Course Section:1				

L_32_624_986_L_1505_41467



Name of the candidate. Eeshan Patrick Gurus

Candidate's UID number:

2	0	2	2	7	0	0	0	1	1
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Examination class room number:

4	0	9
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Invigilator's signature with date:

 08/10/25

(To be filled in by the candidate)

EXAMINATION: BE BTECH
(For example FY MCA/FY BTECH)

SEMESTER :

PROGRAM: CSE-DS

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: Wednesday 8/10/25

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Q1 a)

Ans.

~~Ans.~~

we know that, gradient descent weight update formula =

$$w_{new} = w_{old} - n \cdot \frac{\partial f(w)}{\partial w}$$

$$\frac{\partial f(w)}{\partial w} = \frac{\partial}{\partial w} \left[\frac{1}{2} \|Xw - y\|^2 \right]$$

$$= \frac{\partial}{\partial w} \left[\frac{1}{2} [X^2 w^2 + y^2 - 2Xwy] \right]$$

$$= \frac{1}{2} [2X^2 w - 2Xy]$$

~~$$= X^2 w - Xy$$~~

$$= X^2 w - Xy$$

$$\therefore w_{new} = w_{old} - n(X^2 w - Xy)$$

let λ_{max} = largest eigenvalue of $X^T X$

$$\text{let } n = \frac{2}{\lambda_{max}}$$

$$w_{new} = w - nX^2 w + nXy$$

$$= w - \frac{2}{\lambda_{max}} X^2 w + \frac{2}{\lambda_{max}} Xy$$



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	Q1b)	Total Error = Bias² + Variance + Noise When model capacity is too high, a large number of variables are used to train the model. Due to this, the model tries to learn weights that creates a best-fit line that is highly accurate for the training data due to the large number of variables. The model bias increases due to the no. of variables in the input data and this causes causes overfitting as the model cannot make accurate predictions on the test data due to its high bias to the training data.

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~~Q2a~~

Q2a

A	B	C	D	E
F	G	H	I	J
K	L	M	N	O
P	Q	R	S	T
U	V	W	X	Y

Input (HxWxK)

a	b	c
d	e	f
g	h	i

Filter/Kernel
(FxF)

Let $H=W=5$, $C=3$
 $F=3$, $k=2$

After padding,

Filter
Kernel

1	1	1			
1	A	B	C	D	E
1	F	G	H	I	J
	K	L	M	N	O
	P	Q	R	S	T
	U	V	W	X	Y

Output at cell A

$$= a+b+c+d+e+ \\ b+f+g+h+i$$

For ~~one~~ convolution operation, $F \times F$ multiplications

~~For HxW~~Across W, multiplications = $W \times F \times F$ Across H and W, multiplications = $H \times W \times F \times F$

Across Channels C, multiplication = ~~$H \times W \times C \times F \times F$~~
 $H \times W \times C \times F \times F$

~~Adding~~

Across multiple filters $k = H \times W \times C \times F \times F \times k$

But for a standard 2D convolution,

A	B	C	D	E
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Input length = 5

X	Y	Z
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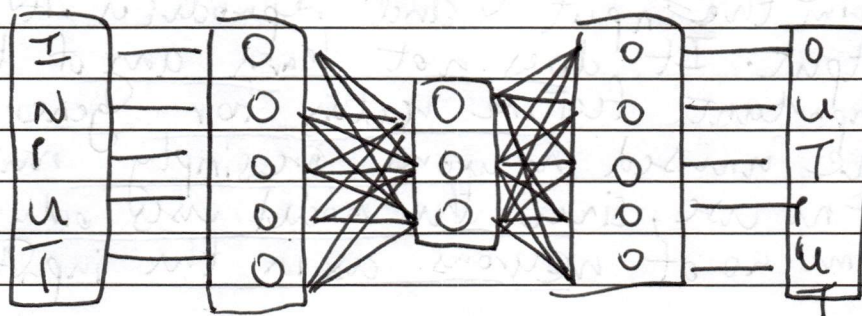
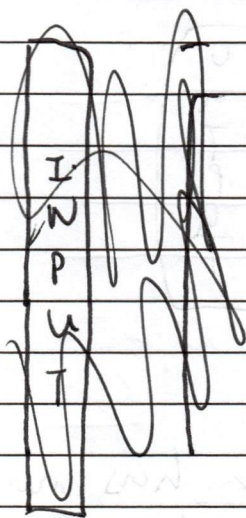


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		If input size size = n filter size = k <u>Max No. of Multiplications = $n \times k$</u> Depthwise separable convolution are critical in modern architectures to capture patterns across channels. Convolution is a technique used to capture information and pattern in the data. However, if we want to capture these patterns across channels (R, G, B), we will need to use depthwise convolution. They are critical for capturing image information across multiple frequency channels. Parameter efficiency :- 1) Depthwise convolutions ensure parameter efficiency by making the parameters learn weights across multiple channels instead of a single channel. 2) The parameters learn weights differently across each channel. If a single channel is used, the parameters won't be able to efficiently learn weights.

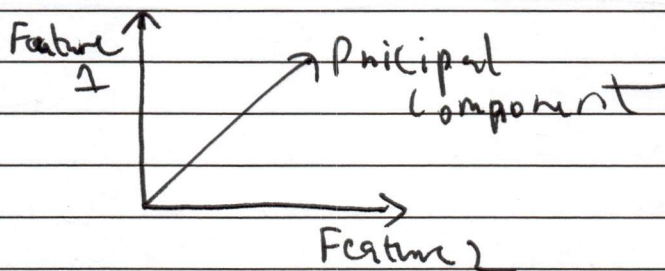
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(Q26) Linear Under complete autoencoder:-



In PCA, we merge two or more features along the principal components of their axes.



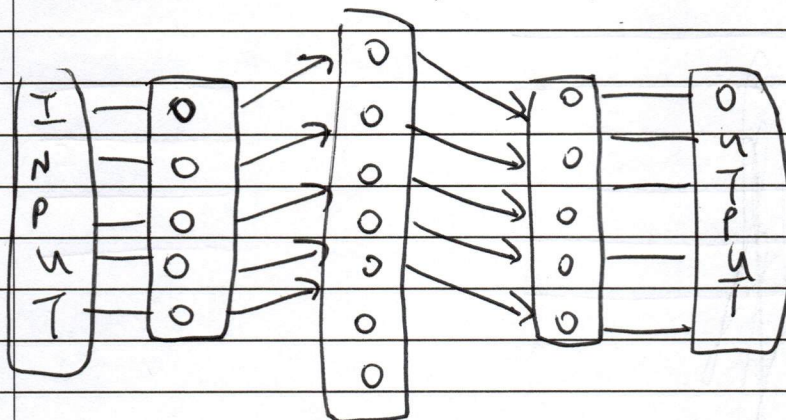
In ~~auto~~ undercomplete autoencoder, we ~~for~~ force the hidden layer to learn the most important features and generalizations of the input channels in a reduced subspace compared to the input and output. It minimizes the error of output and input. Therefore, undercomplete autoencoder when trained with $MSE = |\hat{y} - y|^2$ learns a subspace equivalent to PCA.

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In an overcomplete autoencoder,

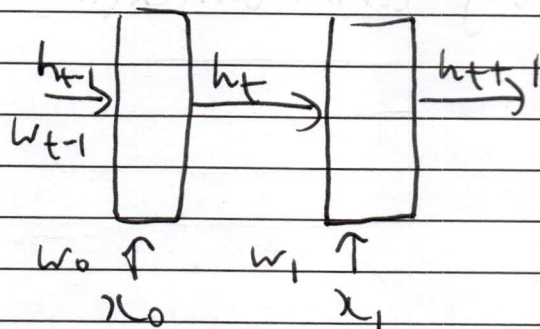


~~The~~ Since the hidden layer has more neurons than the input and output layers, it simply copies the identity function from the input and reproduces it in the output. It does not learn any of the important feature vectors for generalization. The unused neurons are simply null and of no use, since the model uses only the same no of neurons as in the input layer.

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Q3a For an RNN,



In the vanishing gradient problem, the RNN, forgets context after two or 3 words. It can only remember short term context and is unable to recall long term memory.

For an RNN, ~~h_t = \sigma(h_{t-1} \cdot w_{t-1} + x_t \cdot w_0)~~

$$h_t = \sigma(h_{t-k} \cdot w_{t-k} + w_n x_t + b_n)$$

$$y_t = \sigma(h_t \cdot w_t + b_t)$$

$$h_t = \sigma(h_{t-k} \cdot w_{t-k} + w_t \cdot x_t + b_t)$$

$$y_t = \sigma(h_t \cdot w_t^y + b_t)$$

In an LSTM, ~~the~~ it makes use of 3 gates, forget, input and output. The forget gate decides what to discard, the input handles the current input, the cell state takes care of long term context and the output gate produces the output.

Its equations are:-

$$\text{Forget: } h_f = \sigma[W(h_{f-1}, x_{f-1}) + b_{f-1}]$$

$$\text{Input: } h_i = \sigma[W(h_{i-1}, x_{i-1}) + b_{i-1}]$$



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$$\text{Output} = h_0 = \sigma(W(h_{0-1}, x_{0-1}) + b_{0-1})$$

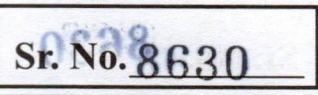
$\therefore$ It retains long-term context.

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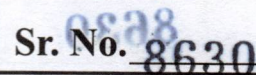
Q36

GRUs tend to have fewer parameters than LSTMs since they combine the forget and input gate to the update gate. ~~and~~

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]